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3.5 Reference Sheets

Arithmetic and Decimals

ABSOLUTE VALUE:

$|x|$ is x if $x \geq 0$ and $-x$ if $x < 0$.

For any x and y, $|x+y| < |x| + |y|$, and $|xy| = |x||y|$.

$\sqrt{x^2} = |x|$.

EVEN AND ODD NUMBERS:

Even × Even = Even	Even × Odd = Even
Odd × Odd = Odd	Even + Even = Even
Even + Odd = Odd	Odd + Odd = Even

ADDITION AND SUBTRACTION:

$x + 0 = x = x - 0$

$x - x = 0$

$x + y = y + x$

$x - y = -(y - x) = -y + x$

$(x + y) + z = x + (y + z)$

If x and y are both positive, then $x + y$ is positive.

If x and y are both negative, then $x + y$ is negative.

DECIMALS WITH MANY FINITE DIGITS:

Add or subtract decimals by lining up their decimal points:

17.6512	653.2700
+ 653.2700	−17.6512
670.9212	635.6188

To multiply decimal A by decimal B:

First, ignore the decimal points and multiply A and B as if they were integers.

Next, if decimal A has n digits to the right of its decimal point and decimal B has m digits to the right of its decimal point, place the decimal point in $A \times B$ so it has $m + n$ digits to its right.

To divide decimal A by decimal B, first move the decimal points of A and B equally many digits to the right until B is an integer. Then divide as you would

QUOTIENTS AND REMAINDERS:

The quotient q and the remainder r of dividing positive integer x by positive integer y are unique positive integers such that

$y = xq + r$ and $0 < r < x$.

The remainder r is 0 if and only if y is divisible by x. Then x is a factor of y.

MULTIPLICATION AND DIVISION:

$x \times 1 = x = \frac{x}{1}$

$x \times 0 = 0$

If $x \neq 0$, then $\frac{x}{x} = 1$.

$\frac{x}{0}$ is undefined.

$xy = yx$

If $x \neq 0$ and $y \neq 0$, then $\frac{x}{y} = \frac{1}{\left(\frac{y}{x}\right)}$.

$(xy)z = x(yz)$

$xy + xz = x(y + z)$

If $y \neq 0$, then $\left(\frac{x}{y}\right) + \left(\frac{z}{y}\right) = \frac{(x+z)}{y}$.

If x and y are both positive, then xy is positive.

If x and y are both negative, then xy is positive.

If x is positive and y is negative, then xy is negative.

If $xy = 0$, then $x = 0$ or $y = 0$, or both.

SCIENTIFIC NOTATION:

To convert a number in scientific notation $A \times 10^n$ into regular decimal notation, move A's decimal point n places to the right if n is positive, or $|n|$ places to the left if n is negative.

To convert a decimal to scientific notation, move the decimal point n spaces so that exactly one nonzero digit is to its left. Multiply the result by 10^n if you moved the decimal point to the left or by 10^{-n} if you moved it to the right.

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right until B is an integer. Then divide as you would integers.

ARITHMETIC SHORTCUTS:

For any two nonnegative integers x and y, the final digit of $x + y$ is the final digit of the sum of x's final digit and y's final digit. And for any two integers x and y, the final digit of xy is the final digit of the product of x's final digit and y's final digit.

An integer is divisible by	if and only if
2	its final digit is 0, 2, 4, 6, or 8.
3	the sum of its digits is divisible by 3.
4	its final two digits make a number divisible by 4.
5	its final digit is 5 or 0.
9	the sum of its digits is divisible by 9.
10	its final digit is 0.

Exponents

SQUARES, CUBES, AND SQUARE ROOTS:

Every positive number has two real square roots, one positive and the other negative. The table below shows the positive square roots rounded to the nearest hundredth.

n	n^2	n^3	$\sqrt{n}$
1	1	1	1
2	4	8	1.41
3	9	27	1.73
4	16	64	2
5	25	125	2.24
6	36	216	2.45
7	49	343	2.65
8	64	512	2.83
9	81	729	3
10	100	1,000	3.16

EXPONENTIATION:

Formula	Example
$x^1 = x$	$2^1 = 2$
If $x \neq 0$, then $x^0 = 1$.	$2^0 = 1$
If $x \neq 0$, then $x^{-1} = \frac{1}{x}$.	$2^{-1} = \frac{1}{2}$
If $x > 1$ and $y > 1$, then $x^y > x$.	$2^3 = 8 > 2$
If $0 < x < 1$ and $y > 1$, then $x^y < x$.	$0.2^3 = 0.008 < 0.2$
$(x^y)^z = x^{yz} = (x^z)^y$	$(2^3)^4 = 2^{12} = (2^4)^3$
$x^{y+z} = x^y x^z$	$2^7 = 2^3 2^4$
If $x \neq 0$, then $x^{y-z} = \frac{x^y}{x^z}$.	$2^{5-3} = \frac{2^5}{2^3}$

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10	100	1,000	3.10

$(xz)^y = x^y z^y$	$6^4 = 2^4 3^4$
If $z \neq 0$, then $\left(\frac{x}{z}\right)^y = \frac{x^y}{z^y}$.	$\left(\frac{3}{4}\right)^2 = \frac{3^2}{4^2} = \frac{9}{16}$
If $z \neq 0$, then $x^{\frac{y}{z}} = (x^y)^{\frac{1}{z}} = (x^{\frac{1}{z}})^y$.	$4^{\frac{2}{3}} = (4^2)^{\frac{1}{3}} = (4^{\frac{1}{3}})^2$

Algebraic Expressions and Linear Equations

TRANSLATING WORDS INTO MATH OPERATIONS:

$x+y$	$x-y$	xy	$\frac{x}{y}$	x^y
x added to y x increased by y x more than y x plus y the sum of x and y the total of x and y	x decreased by y difference of x and y y fewer than x y less than x x minus y x reduced by y y subtracted from x	x multiplied by y the product of x and y x times y	x divided by y x over y the quotient of x and y the ratio of x to y	x to the power of y x to the yth power
		If $y = 2$: double x twice x	If $y = 2$: half of x x halved	If $y = 2$: x squared
		If $y = 3$: triple x		If $y = 3$: x cubed

MANIPULATING ALGEBRAIC EXPRESSIONS:

Technique	**Example**
Factor to combine like terms	$3xy - 9y = 3y(x-3)$
Divide out common factors	$\frac{(3xy-9y)}{(x-3)} = \frac{3y(x-3)}{(x-3)} = 3y(1) = 3y$
Multiply two expressions by multiplying each term of one expression by each term of the other	$\begin{aligned}(3x-4)(9y+x) &= 3x(9y+x) - 4(9y+x) \\ &= 3x(9y) + 3x(x) - 4(9y) - 4(x) \\ &= 27xy + 3x^2 - 36y - 4x\end{aligned}$
Substitute constants for variables	If $x = 3$ and $y = -2$, then $3xy - x^2 + y = 3(3)(-2) - (3)^2 + (-2) = -18 - 9 - 2 = -29.$

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SOLVING LINEAR EQUATIONS:

Technique	**Example**
Isolate a variable on one side of an equation by doing the same operations on both sides of the equation.	Solve the equation $\frac{(5x-6)}{3} = 4$ like this: (1) Multiply both sides by 3 to get $5x - 6 = 12$. (2) Add 6 to both sides to get $5x = 18$. (3) Divide both sides by 5 to get $x = \frac{18}{5}$.
To solve two equations with two variables x and y: (1) Express x in terms of y using one of the equations. (2) Substitute that expression for x to make the second equation have only the variable y. (3) Solve the second equation for y. (4) Substitute the solution for y into the first equation to solve for x.	Solve the equations A: $x - y = 2$ and B: $3x + 2y = 11$: (1) From A, $x = 2 + y$. (2) In B, substitute $2 + y$ for x to get $3(2+y) + 2y = 11$. (3) Solve B for y: $6 + 3y + 2y = 11$ $\begin{aligned} 6 + 5y &= 11 \\ 5y &= 5 \\ y &= 1 \end{aligned}$ (4) Since $y = 1$, from A we find $x = 2 + 1 = 3$.
Alternative technique: (1) Multiply both sides of one equation or both equations so that the coefficients on y have the same absolute value in both equations. (2) Add or subtract the two equations to remove y and solve for x. (3) Substitute the solution for x into the first equation to find the value of y.	Solve the equations A: $x - y = 2$ and B: $3x + 2y = 11$: (1) Multiply both sides of A by 2 to get $2x - 2y = 4$. (2) Add the equation in (1) to equation B: $\begin{aligned} 2x - 2y + 3x + 2y &= 4 + 11 \\ 5x &= 15 \\ x &= 3 \end{aligned}$ (3) Since $x = 3$, from A we find $3 - y = 2$, so $y = 1$.

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Factoring, Quadratic Equations, and Inequalities

SOLVING EQUATIONS BY FACTORING:

Techniques	**Example**
(1) Start with a polynomial equation. (2) Add or subtract from both sides until 0 is on one side of the equation. (3) Write the nonzero side as a product of factors.	$\begin{aligned} x^3 - 2x^2 + x &= -5(x-1)^2 \\ x^3 - 2x^2 + x + 5(x-1)^2 &= 0 \\ \text{(i)} \quad x(x^2 - 2x + 1) + 5(x-1)^2 &= 0 \\ \text{(ii)} \quad x(x-1)^2 + 5(x-1)^2 &= 0 \\ \text{(iii)} \quad (x+5)(x-1)^2 &= 0 \end{aligned}$ $x + 5 = 0$ or $x - 1 = 0$. So $x = -5$ or $x = 1$

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(4) Set each factor to 0 to find simple equations giving the solutions to the original equation.	$x+5=0$ or $x-1=0$. So, $x=-5$ or $x=1$.

FORMULAS FOR FACTORING:

$$\left(a+\frac{1}{a}\right)^2 = a^2+\frac{1}{a^2}+2$$

$$a^2-b^2=(a-b)(a+b)$$

$$a^2+2ab+b^2=(a+b)^2$$

$$a^2-2ab+b^2=(a-b)^2$$

$$a^2+b^2=(a+b)^2-2ab=(a-b)^2+2ab$$

$$(a+b+c)^2=a^2+b^2+c^2+2ab+2ac+2bc$$

$$(a+b)^3=a^3+3a^2b+3ab^2+b^3$$

THE QUADRATIC FORMULA:

For any quadratic equation $ax^2+bx+c=0$ with $a\neq 0$, the roots are

$$x=\frac{-b+\sqrt{b^2-4ac}}{2a} \text{ and } x=\frac{-b-\sqrt{b^2-4ac}}{2a}$$

These roots are two distinct real numbers if $b^2-4ac\geq 0$.

If $b^2-4ac=0$, the equation has only one root: $-\frac{b}{2a}$.

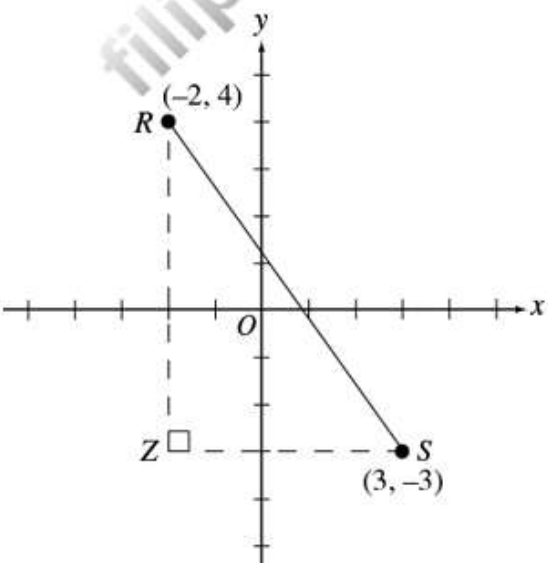
If $b^2-4ac<0$, the equation has no real roots.

SOLVING INEQUALITIES:

Explanation	Example
As in solving an equation, the same number can be added to or subtracted from both sides of the inequality, or both sides can be multiplied or divided by a positive number, without changing the order of the inequality. But multiplying or dividing an inequality by a negative number reverses the order of the inequality. Thus, 6 > 2, but (−1)(6) < (−1)(2).	To solve the inequality $\frac{(5x-1)}{-2}<3$ for x, isolate x: (1) $5x-1>-6$ (multiplying both sides by −2, reversing the order of the inequality) (2) $5x>-5$ (add 1 to both sides) (3) $x>-1$ (divide both sides by 5)

LINES IN THE COORDINATE PLANE:

An equation $y=mx+b$ defines a line with slope m which has a y-intercept that is b.



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For a line through two points (x_1, y_1) and (x_2, y_2) with $x_1 \neq x_2$, the slope is $m = \dfrac{(y_2 - y_1)}{(x_2 - x_1)}$. Given the known point (x_1, y_1) and the slope m, any other point (x, y) on the line satisfies the equation $m = \dfrac{(y - y_1)}{(x - x_1)}$.

Above, the line's slope is $\dfrac{(-3-4)}{(3-(-2))} = -\dfrac{7}{5}$. To find an equation of the line, we can use the point (3, −3):

$$\begin{aligned} y-(-3) &= \left(-\frac{7}{5}\right)(x-3) \\ y+3 &= \left(-\frac{7}{5}\right)x + \frac{21}{5} \\ y &= \left(-\frac{7}{5}\right)x + \frac{6}{5} \end{aligned}$$

So, the y-intercept is $\dfrac{6}{5}$.

Find the x-intercept like this:

$$\begin{aligned} 0 &= \left(-\frac{7}{5}\right)x + \frac{6}{5} \\ \left(\frac{7}{5}\right)x &= \frac{6}{5} \\ x &= \frac{6}{7} \end{aligned}$$

The graph shows both these intercepts.

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Rates, Ratios, and Percents

FRACTIONS:

Equivalent or Equal Fractions:

Two fractions stand for the same number if dividing each fraction's numerator and denominator by their greatest common divisor makes the fractions identical.

Adding, Subtracting, Multiplying, and Dividing Fractions:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad}{bd} + \frac{bc}{bd}; \frac{a}{b} - \frac{c}{d} = \frac{ad}{bd} - \frac{bc}{bd}$$

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}; \frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc}$$

MIXED NUMBERS:

A mixed number of the form $a\dfrac{b}{c}$ equals the fraction $\dfrac{ac+b}{c}$.

PERCENTS:

$x\% = \dfrac{x}{100}$.

$x\%$ of y equals $\dfrac{xy}{100}$.

To convert a percent to a decimal, drop the percent sign, then move the decimal point two digits left.

To convert a decimal to a percent, add a percent sign, then move the decimal point two digits right.

PERCENT INCREASE OR DECREASE:

The percent increase from x to y is $100\left(\dfrac{y-x}{x}\right)\%$.

The percent decrease from x to y is $100\left(\dfrac{x-y}{x}\right)\%$.

DISCOUNTS:

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RATE:

distance = rate × time

PROFIT:

Profit = Revenues − Expenses, or

Profit = Selling price − Cost.

INTEREST:

Simple annual interest = (principal) × (interest rate) × (time)

Compound interest over n periods = (principal) × $(1 + \text{interest per period})^n$ − principal

DISCOUNTS:

A price discounted by n percent becomes $(100 - n)$ percent of the original price.

A price discounted by n percent and then by m percent becomes $(100 - n)(100 - m)$ percent of the original price.

WORK:

$\frac{1}{r} + \frac{1}{s} = \frac{1}{h}$, where r is how long one individual takes to do an amount of work, s is how long a second individual takes to do that much work, and h is how long they take to do that much work when both are working at the same time.

MIXTURES:

	Number of units of a substance or mixture	Amount of an ingredient per unit of the substance or mixture	Total amount of that ingredient in the substance or mixture
Substance A	X	M	X × M
Substance B	Y	N	Y × N
Mixture of A and B	X + Y	$\frac{(X \times M) + (Y \times N)}{X + Y}$	(X × M) + (Y × N)

Statistics, Sets, and Counting Methods

STATISTICS:

Concept	**Definition for a data set of n values ordered from least to greatest**	**Example with data set {4, 4, 5, 7, 10}**
Mean	The sum of the n values, divided by n	$\frac{(4+4+5+7+10)}{5} = \frac{30}{5} = 6$
Median	The middle value if n is odd; The mean of the two middle values if n is even.	5 is the middle value in {4, 4, 5, 7, 10}.
Mode	The value that appears most often in the data set.	4 is the only value that appears more than once in {4, 4, 5, 7, 10}.
Range	The largest value in the data set minus the smallest	10 − 4 = 6
Standard Deviation	Calculate like this: (1) Find the arithmetic mean (2) Find the differences between each of the n	(1) The mean is 6. (2) −2, −2, −1, 1, 4 (3) 4, 4, 1, 1, 16

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	values and the mean (3) Square each of the differences (4) Find the average of the squared differences, and (5) Take the nonnegative square root of this average	(4) $\frac{26}{5} = 5.2$ (5) $\sqrt{5.2}$

SETS:

Concept	Notation for finite sets S and T	Example
Number of elements	$\lvert S\rvert$	$S = \{-5, 0, 1\}$ is a set with $\lvert S\rvert = 3$.
Subset	$S \subseteq T$ (S is a subset of T); $S \supseteq T$ (T is a subset of S)	$\{-5, 0, 1\}$ is a subset of $\{-5, 0, 1, 4, 10\}$.
Union	$S \cup T$	$\{3, 4\} \cup \{4, 5, 6\} = \{3, 4, 5, 6\}$
Intersection	$S \cap T$	$\{3, 4\} \cap \{4, 5, 6\} = \{4\}$
The general addition rule for two sets	$\lvert S \cup \mathrm{T}\rvert = \lvert S\rvert + \lvert \mathrm{T}\rvert - \lvert S \cap \mathrm{T}\rvert$	$\lvert\{3, 4\} \cup \{4, 5, 6\}\rvert =$ $\lvert\{3, 4\}\rvert + \lvert\{4, 5, 6\}\rvert - \lvert\{3, 4\} \cap \{4, 5, 6\}\rvert =$ $\lvert\{3, 4\}\rvert + \lvert\{4, 5, 6\}\rvert - \lvert\{4\}\rvert = 2 + 3 - 1 = 4.$

COUNTING METHODS:

Concept and Equations	Examples
Multiplication Principle: The number of possible choices of 1 element apiece from the data sets $A_1, A_2, \ldots, A_n$ is $\lvert A_1\rvert \times \lvert A_2\rvert \times \ldots \times \lvert A_n\rvert$.	The number of possible choices of 1 element apiece from the data sets $S = \{-5, 0, 1\}$, $T = \{3, 4\}$, and $U = \{3, 4, 5, 6\}$ is $\lvert S\rvert \times \lvert T\rvert \times \lvert U\rvert = 3 \times 2 \times 4 = 24.$
Factorial: $n! = n \times (n-1) \times \ldots \times 1$ $0! = 1! = 1$ $n! = (n-1)!(n)$	4! = 4 × 3 × 2 × 1 = 24 4! = 3! × 4
Permutations: A data set of n objects has $n!$ permutations	The data set of letters A, B, and C has 3! = 6 permutations: ABC, ACB, BAC, BCA, CAB, and CBA.
Combinations: The number of possible choices of k objects from a data set of n objects is $\binom{n}{k} = \frac{n!}{k!(n-k)!}$.	The number of 2-element subsets of data set {A, B, C, D, E} is $\binom{5}{2} = \frac{5!}{2!3!} = \frac{120}{(2)(6)} = 10.$

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